This assignment will not be graded and is only for practice.

Level 1:

1) Consider a system of linear equations as follows: ***1 point***

$$a_{11}x_1 + a_{12}x_2 + a_{13}x_3 = b_1$$
$$a_{21}x_1 + a_{22}x_2 + a_{23}x_3 = b_2$$
$$a_{31}x_1 + a_{32}x_2 + a_{33}x_3 = b_3$$

Let the matrix representation of the above system be $Ax = b$, where $A = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}$, $x = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$, and $b = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix}$. Let A_{x_i} be the matrix obtained by replacing the i-th column of A $(i.e., \begin{bmatrix} a_{1i} \\ a_{2i} \\ a_{3i} \end{bmatrix}$)by b, for $i = 1, 2, 3$. If $det(A_{x_i}) = 0$ for $i = 1, 2, 3$, then which of the following is (are) true?

- ☐ $x = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$ must be a solution.
- ☐ $x = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$ is a solution if $det(A) = 0$.
- ☐ If $det(A) = 0$, then we cannot conclude about the solution using Cramer's rule.
- ☐ If $det(A) = 0$, then $b = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$.

No, the answer is incorrect.

Score: 0

Accepted Answers:

$x = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$ is a solution if $det(A) = 0$.

If $det(A) = 0$, then we cannot conclude about the solution using Cramer's rule.

If $det(A) = 0$, then $b = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$.

Consider a system of linear equations

$$\begin{aligned} x_1 + x_3 &= 1 \\ -x_1 + x_2 - x_3 &= 1 \\ -x_2 + x_3 &= 1 \end{aligned}$$

Let matrix representation of the above system be $Ax = b$, where $A = \begin{bmatrix} 1 & 0 & 1 \\ -1 & 1 & -1 \\ 0 & -1 & 1 \end{bmatrix}$, $x = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$, and $b = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$. Let A_{x_i} be the matrix obtained by replacing the i-th column of A (i.e., $\begin{bmatrix} a_{1i} \\ a_{2i} \\ a_{3i} \end{bmatrix}$) by b, for $i = 1, 2, 3$.

(Use the above information to answer questions 2 and 3)

2) Choose the set of correct options **1 point**

- ☐ $A_{x_1} = \begin{bmatrix} 1 & 1 & 0 \\ -1 & 1 & -1 \\ 0 & -1 & 1 \end{bmatrix}$
- ☐ $A_{x_1} = \begin{bmatrix} 1 & 0 & 1 \\ 1 & 1 & -1 \\ 1 & -1 & 1 \end{bmatrix}$
- ☐ $A_{x_2} = \begin{bmatrix} 1 & 0 & 1 \\ -1 & 1 & -1 \\ 0 & -1 & 1 \end{bmatrix}$
- ☐ $A_{x_3} = \begin{bmatrix} 1 & 0 & 1 \\ -1 & 1 & 1 \\ 0 & -1 & 1 \end{bmatrix}$

No, the answer is incorrect.

Score: 0

Accepted Answers:

$A_{x_1} = \begin{bmatrix} 1 & 0 & 1 \\ 1 & 1 & -1 \\ 1 & -1 & 1 \end{bmatrix}$

$A_{x_3} = \begin{bmatrix} 1 & 0 & 1 \\ -1 & 1 & 1 \\ 0 & -1 & 1 \end{bmatrix}$

3) Choose the set of correct options. **1 point**

- ☐ $x_1 = -2.$
- ☐ $x_2 = -2.$
- ☐ $x_3 = 3.$
- ☐ None of the above.

No, the answer is incorrect.

Score: 0

Accepted Answers:

$x_1 = -2.$
$x_3 = 3.$

Level 2:

4)
Consider the system of linear equations $Ax = b$, where $A = \begin{bmatrix} 1 & a & 0 \\ a & 1 & 2 \\ 1 & 0 & a \end{bmatrix}$, $x = \begin{bmatrix} x_1 \\ 0 \\ x_3 \end{bmatrix}$, and $b = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$, and the solution for x is partially known. What is the value of a^2, if $a = 0, \sqrt{3}, -\sqrt{3}$ is given? [Hint: Observe that the second row of the vector x is given as 0, which implies that x_2 is known, where $x = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$

No, the answer is incorrect.

Score: 0

Accepted Answers:

(Type: Numeric) 2

1 point

5) Consider the system of linear equations $Ax = b$, where $A = \begin{bmatrix} 1 & 2a+3 \\ 3a+2 & 1 \end{bmatrix}$, $x = \begin{bmatrix} x_1 \\ 0 \end{bmatrix}$, and $b = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$, and the solution for x is partially known.

What is the value of $3a$, if $det(A) = 0$ is given?

No, the answer is incorrect.

Score: 0

Accepted Answers:

(Type: Numeric) -1

1 point

Your score is: 0/5